

# Un-Obtrusively Monitoring

*Changing the way how we monitor Neonatal Patients !*

## Overview Healthcare

**Arindam Mukherjee**  
on behalf of Luxisens Team

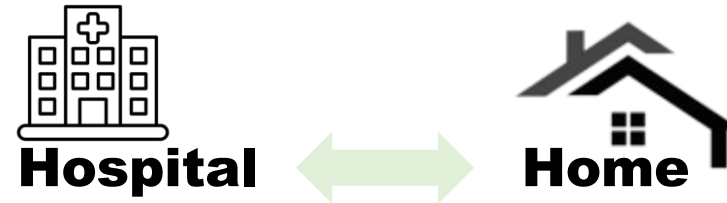


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## Our Vision

# *‘Unobtrusive Monitoring’*



*contactless solution  
to measure clinical vital parameters  
across care settings !*

# Overview

## Key Propositions



**PATENTED OPTICAL TECH INNOVATION**  
REVOLUTIONARY SENSORMATS  
FOR UNOBTRUSIVE MONITORING



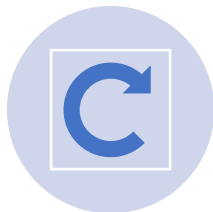
**MARKET OPPORTUNITY**  
HIGH ACROSS VARIOUS  
CLINICAL CARE SETTING



**IMPROVED CLINICAL OUTCOMES:**  
UNOBTRUSIVE, CONTINUOUS  
MEASUREMENT & ACCURACY [1<sup>ST</sup> VITAL  
PARAMETER]



**IMPROVED ECONOMIC OUTCOMES:**  
LESSENING STAFF MONITORING TIME IN  
NICU & OTHER SETTINGS, ESTIMATE  
>20% LOWER COST OF CARE.



**PATENTS GRANTED**  
3  
[1 PENDING]



**TEAM**  
HIGHLY EXPERIENCED  
MANAGEMENT TEAM

Based in Eindhoven  
The Netherlands

Spin-Out of



Our Clinical Partners



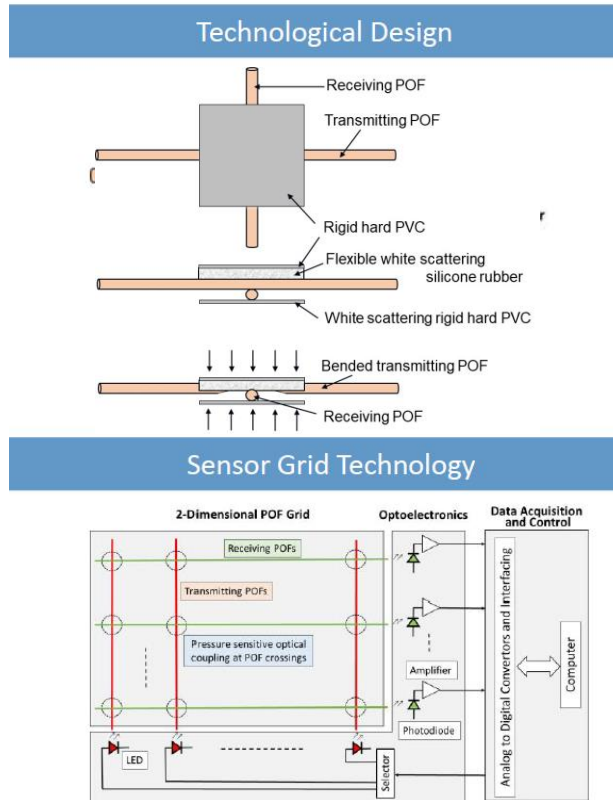
Our Investor



# Technology

## Breakthrough in hardware and software

### 1. Excellent / Smart usage of Optical Fibers



### 2. High Sensitivity Output Pressure

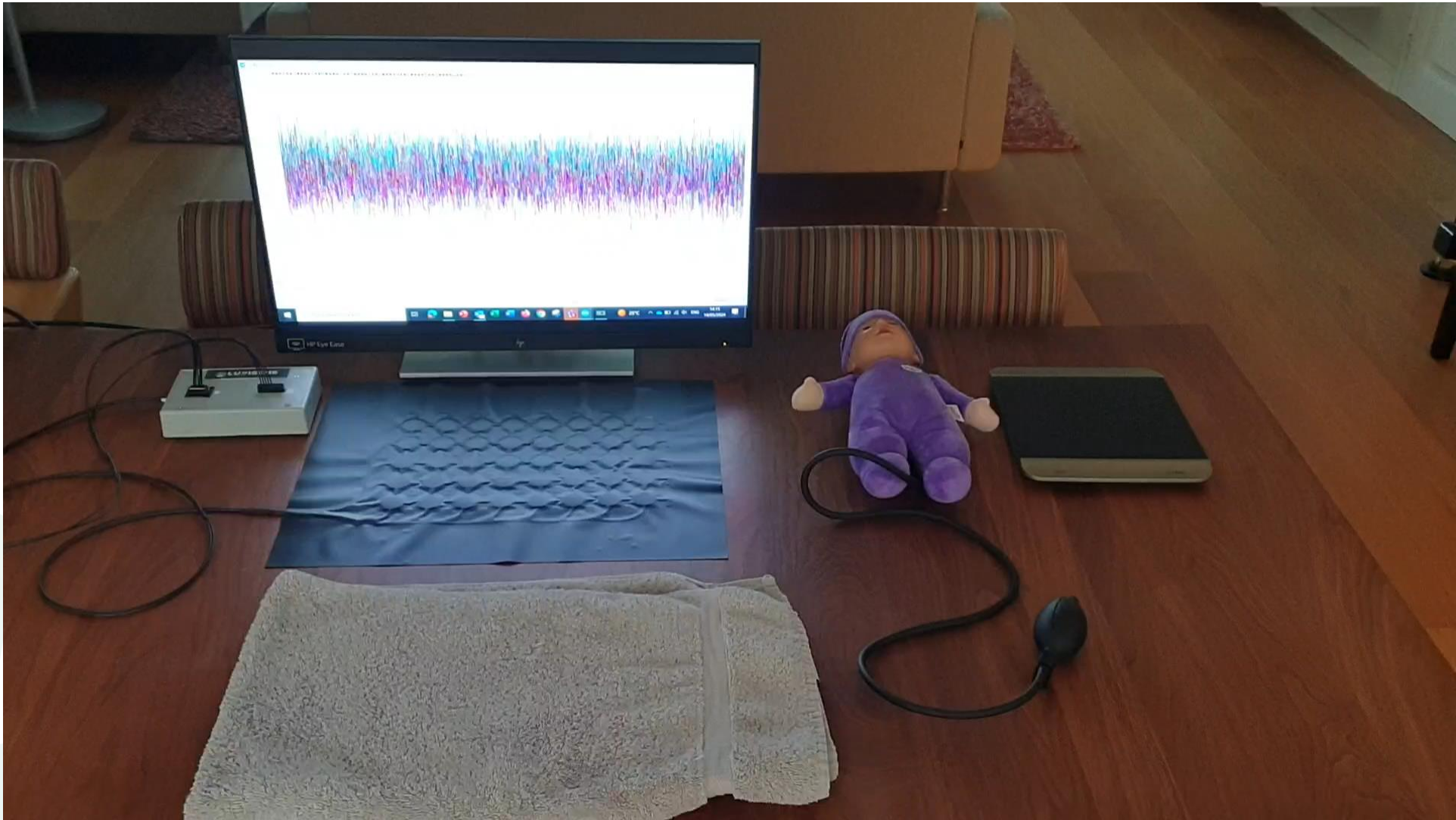


### 3. Secured Supply Chain

- Luxisens usage of components are off-the shelf and are easily available.
- COGs are low from Optical Fiber to Class A components.

New sensor + unique validated software,  
supported by **unique IP**

# Demo





# Broad Go 2 Market (G2M) Segments

Vast potential for end-markets across Industries with current focus on Medical / Healthcare

## Healthcare

- 3 { Bed Mat (care homes)  
Bed Mat (thuiszorg)  
Baby Mat (for home care)  
Floor mat (fall detection)

## Medical

- 1 Neonatal Incubator Mat
- 2 MRI + CT Scan Positioning  
Epilepsy Mat  
Hospital Bed Mat

## Security

- 4 { Warehouse Perimeter  
Fence Perimeter Cable  
Roof and Balcony Mat  
Door Entry Control



## E-Gaming

- Game Mat (support video game)  
Virtual Reality Tools  
Augmented Reality Tools

## Sports

- Field and Athlete Tracking  
Athlete Sleep Analyser

## Industrial

- Windmill Maintenance  
Water Infrastructure  
Construction Safety  
Building Maintenance

# Various clinical segments & opportunities

Focus  
on Top 3



Format – Flat Bed Mats

|                                                         |   |    |   | Current Focus | Procedural Volume | LT Solution Clinical Utility | Standalone / Integrated | Willingness 2 Pay | Regulatory Hurdle | Priority |
|---------------------------------------------------------|---|----|---|---------------|-------------------|------------------------------|-------------------------|-------------------|-------------------|----------|
| NICU [Infants]<br>- Neonatologists                      | H | -- | H | H             | M                 | H                            | SA/Int.                 | H                 | H                 |          |
| Consumer<br>Baby SenseMat [@ Home]                      | H | -- | L | L             | H                 | M                            | SA                      | M                 | L                 |          |
| Elderly Managed Care<br>- Nursing                       | - | H  | H | L             | H                 | NE                           | SA                      | M                 | L                 |          |
| Vital Monitoring<br>- Step Down Units                   |   | H  | H | NP            | H                 | NE                           | Intgd                   | H                 | H                 |          |
| Sleep Diagnosis<br>- Specialist [Sleep, Pulmo..]        |   | H  | H | NP            | H                 | NE                           | SA                      | NE                | NE                |          |
| Decubitus / Palliative Care<br>- Derma / Nursing        |   |    |   |               |                   |                              |                         |                   |                   |          |
| Other Use Cases<br>- Explore [Epileptic, Well Being...] |   |    |   |               |                   |                              |                         |                   |                   |          |
| Imaging [Motion Reduction]<br>- Radiologist             |   |    |   |               |                   |                              |                         |                   |                   |          |

→ Plan for later →

**Research Project**  
1<sup>st</sup> Phase Testing Completed : SenseMat with MRI [+ve Outcomes]

- Value Propositions
- FiH Stakeholders.
- Generate Evidence Case studies.
- Establishing KOL's
- Showcase Clinical Outcomes

CONFIDENTIAL



# The Unmet Need

Improving care management of neonates by measuring vital parameters !



## Care Setting

Neonates with high risk or delicate health conditions are looked after in the NICU (Neonatal Intensive Care Unit) of a hospital. [< 34 weeks are admitted to NICU]

## Clinical Conditions

Continuous monitoring & attention by healthcare professionals is needed for-

- Intrapartum-related complications (**birth asphyxia or inability to breathe at birth**) – is one of the main aspects for neonatal deaths.
- Sepsis prediction using continuous movement is a need.
- Current monitoring is only based on intermittent observation.
- Vital sign data collection to confirm neonate condition [reference of Respiration, possible association with motion and other parameters].

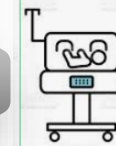
## Challenges

- o delicate skin structures of neonates
- o too many wired connectors
- o pressure on clinical staff for continuous checks
- o intrusive

## Need of the hour:

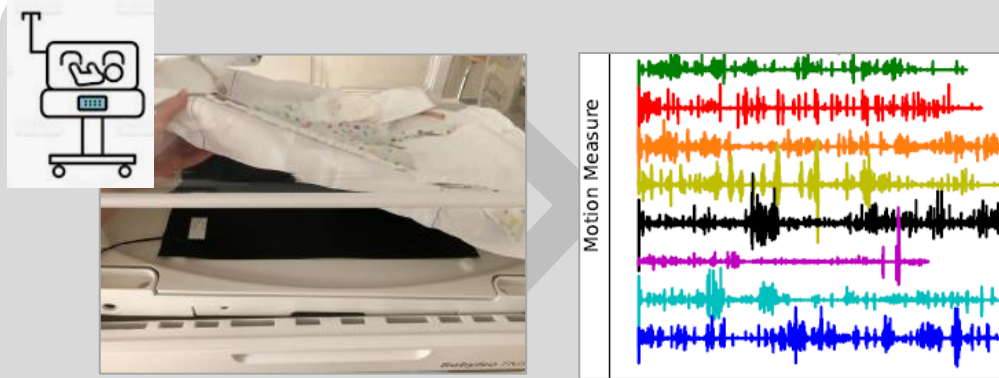
A solution that provides

- active monitoring of neonates.
- in a passive unobtrusive manner.
- in a safe environment
- overcoming the drawbacks of typical vital monitoring systems.



# Collaboration Project 1

Neonatal Mat @ Maxima Medical Center, Veldhoven



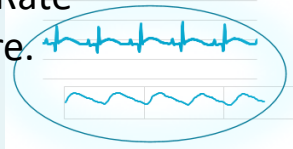
- Used 1<sup>st</sup> version of Luxisens SensorMat.
- Measured in clinical practice
  - Compare fiber mat measurements of motion and respiration with sleep/wake patterns detected using golden standard (PSG)

## Outcomes:

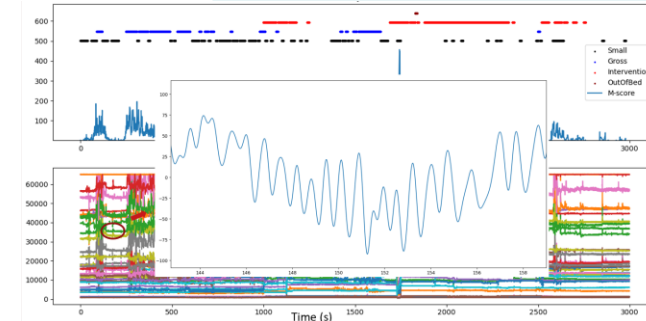
- Clinical paper and results published.
- SensorMat used on 20 babies and yielded high sensitivity & specificity.
- MMC has annotated data on the acquired optical signals.

SensorMat provided

- Positional data
- Respiration Rate
- & many more.



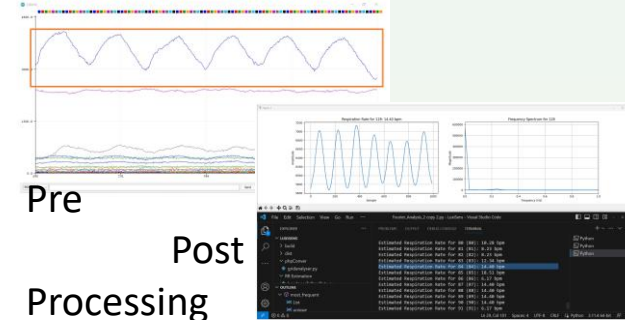
\*initial results



Courtesy / Reference:

Body Motion Detection in Neonates Using a Fiber Mattress by Y. Zhao | Tu/e, MMC, Veldhoven: Guide: Ms. Carola v P.

## Respiration Rate Algorithm Development



Pre

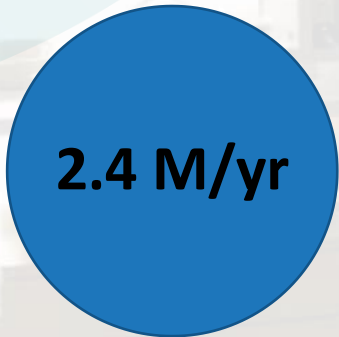
Post

Processing

# Clinical Opportunities & Potential

Monitoring of Neonates in NICU [Neonatal ICU] & beyond !

Neonatal Deaths  
Globally



# NICU Patients  
(Neonates)



% Neonates in  
in NICU



Data:

- Globally: # 140 mln births.
- 3.6 mln births in USA
- 4 mln births in EU

\*source: WHO, NEOIPC Project

## Luxisens SensorMat Solution

**Quickly and accurately** with affordability address the unmet need !

Current competitive solutions require **intrusive patches, intensive physician / staffing attention.**

**A SensorMat for every NICU bed**  
--- potential in huge !

# Potential

## Neonatal Intensive Care Monitoring

- USA Infant Incubator market size: \$ 400 million. CAGR: 6.5%
- Global Infant Incubator market size: \$ 1.2 billion. CAGR: 6.5%
- Working with any of the leading players or selling directly to hospitals with enhanced functionality of Baby SenseMat: will be a good revenue generator.

# Luxisens Value Proposition

Transforming management of patient monitoring at bedside 'Unobtrusively'

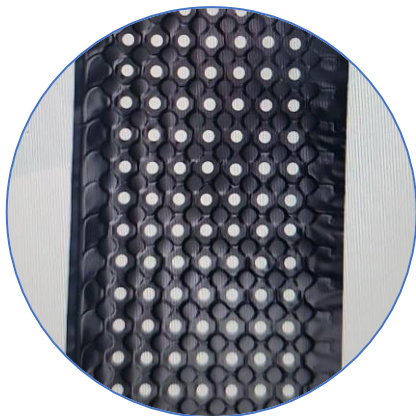


- Patented **Optical fiber technology** in mat format:
  - **Excellent** First in Human **clinical results**.
  - Unique 2D mat solution that provides high clinical accuracy.
- Addressing a large unmet clinical need:
  - **Continuous, accurate, non-invasive measurement of vital parameters across various care settings.**  
**\*Positive** initial 1<sup>st</sup> in Human **clinical results**.
- An impactful solution:
  - Critical when treating **acute setting** (in NICU) & **other vital** monitoring at hospital and at home settings.



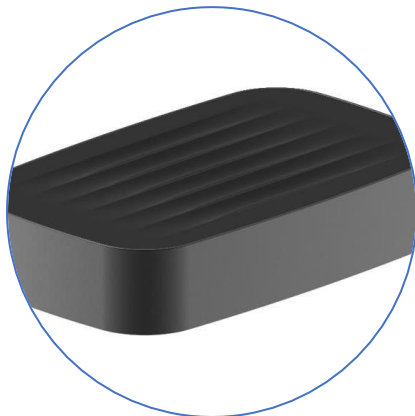
# Luxisens product solution !

Optical sensors with unique software algorithm !



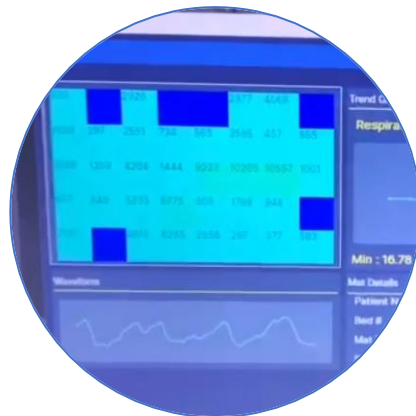
**Optical SenseMat**

various Use Cases



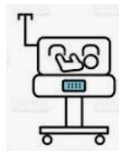
**Advanced Controller**

Wireless / Cloud Connectivity



**Unique Algorithms**

clinically graded



✓ **Unobtrusive**

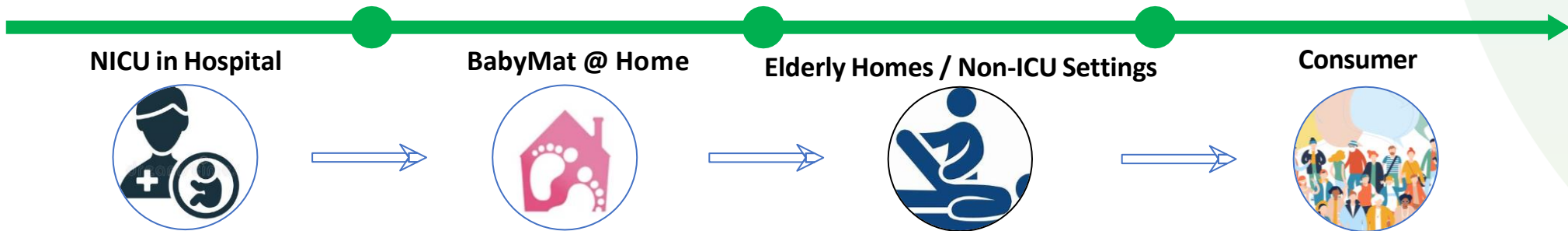
✓ **Affordable**

✓ **Accurate**

✓ **Reliable**

✓ **Operator independent**

## Our journey: Enabling expansion of use cases from hospital to home





# Luxisens product solution !

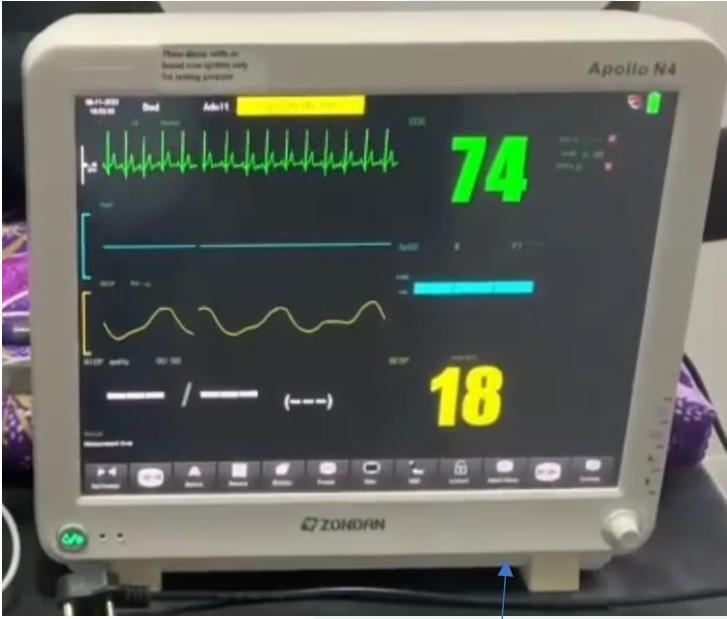
1<sup>st</sup> demonstratable validation of key vital parameter (Respiration Rate)



Luxisens Algorithm Real Time RR Waveform  
Respiration Rate: 18



Eng App: UX/UI under Development  
Respiration Rate: 18



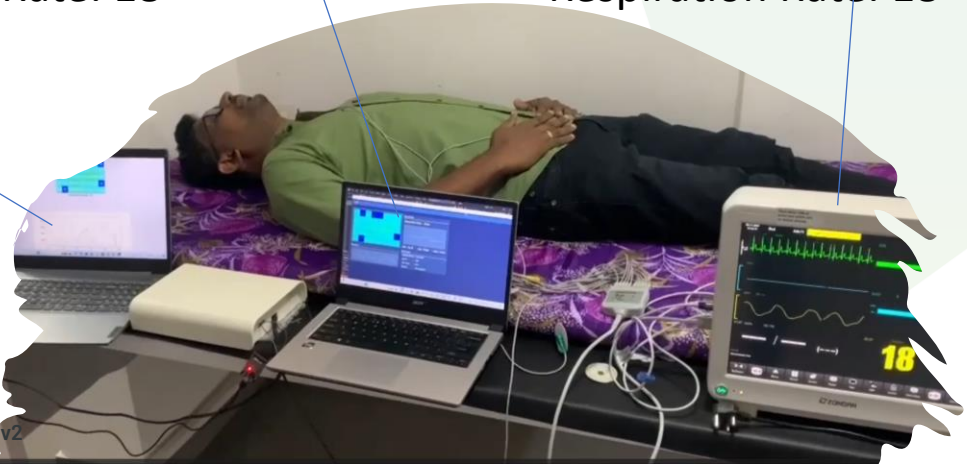
Std Patient Monitor used in Hospitals  
Respiration Rate: 18

## Live Demo [Initial Results] – Proved equivalence !

- Subject on Bed [Thickness: 7+ cms].
- SensorMat version: Small [BabyMat] used.
- Testing done in Supine, Sideways, Flat on Bed.

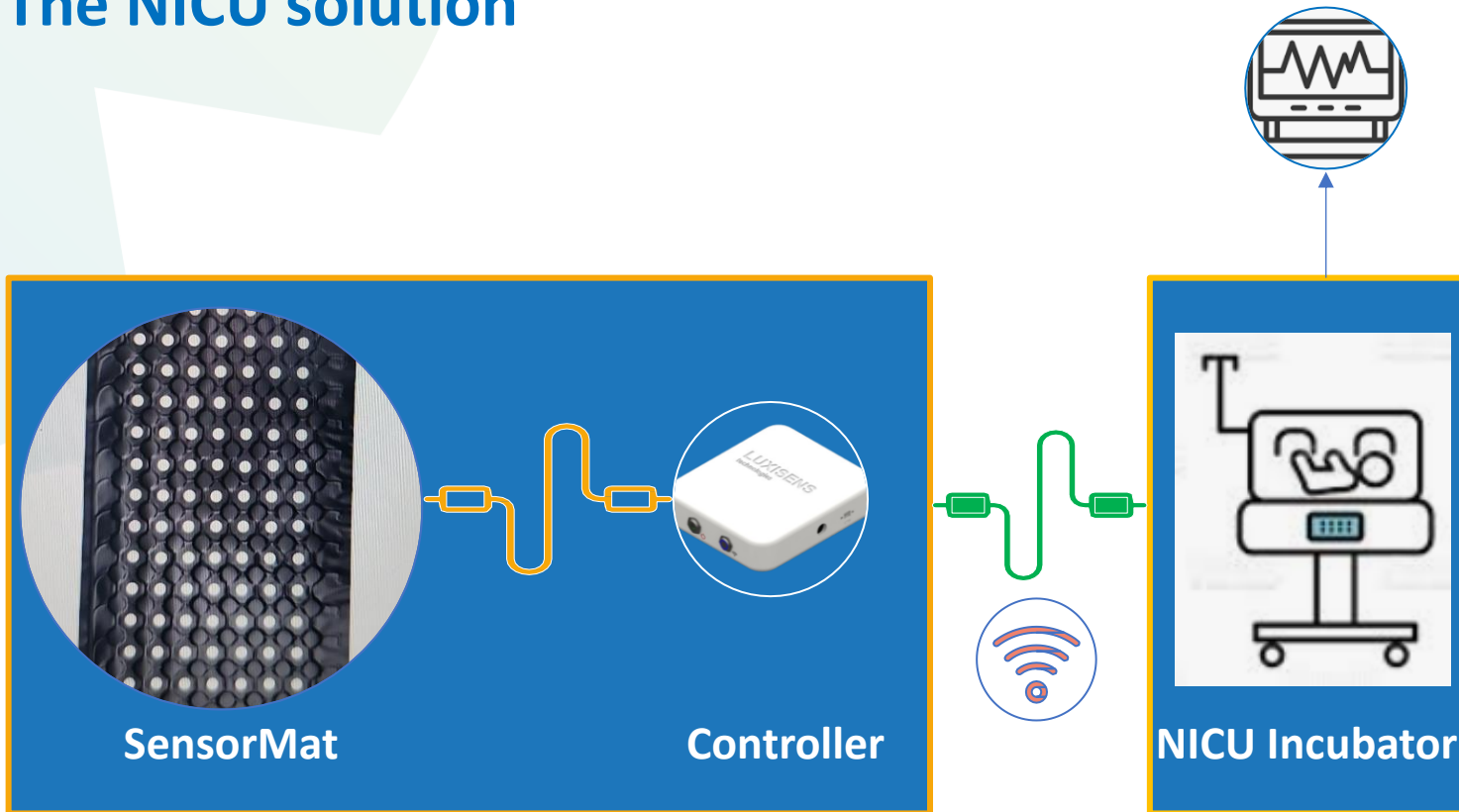
\*Please Note:

This is being demonstrated using still the old controller hardware.  
New h/w Controller + algorithm refinement will further improve results.



# Beachhead Product

## The NICU solution



### Tailored to physician's needs:

- Neonate respiration, motion monitoring.
- Frequency/interval of output measurements to be set by physician.
- Reporting content (data/format) to be set by physician.

### Current Competition:

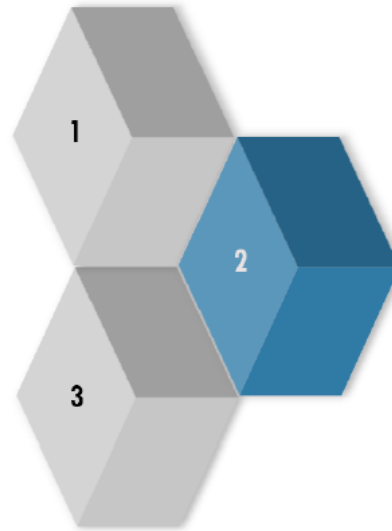
- None [Unobtrusive]

Willingness to Pay : High

# Kickstarting Sales



UNIVERSITY  
OF TWENTE.



## 1. Start: Research Sales

- Sell **Luxisens Research System**
- Develop algorithm business strategy.
- Collaborate / Partner with existing proponents of 'Un-obtrusive' Monitoring.
- Drive Traction with Researchers & also explore with Pharma companies.
- Target audience: Existing researchers who have already published papers in this area.
  - Top: 100+ Hospitals + Researchers
  - Top: 10 Pharma companies [specific clinical domains]
  - End Customer: Sale Value: € 6K– € 7,5K + Value Add Support Services.

## 2. Licensing 'Algorithm (Value Added Services)

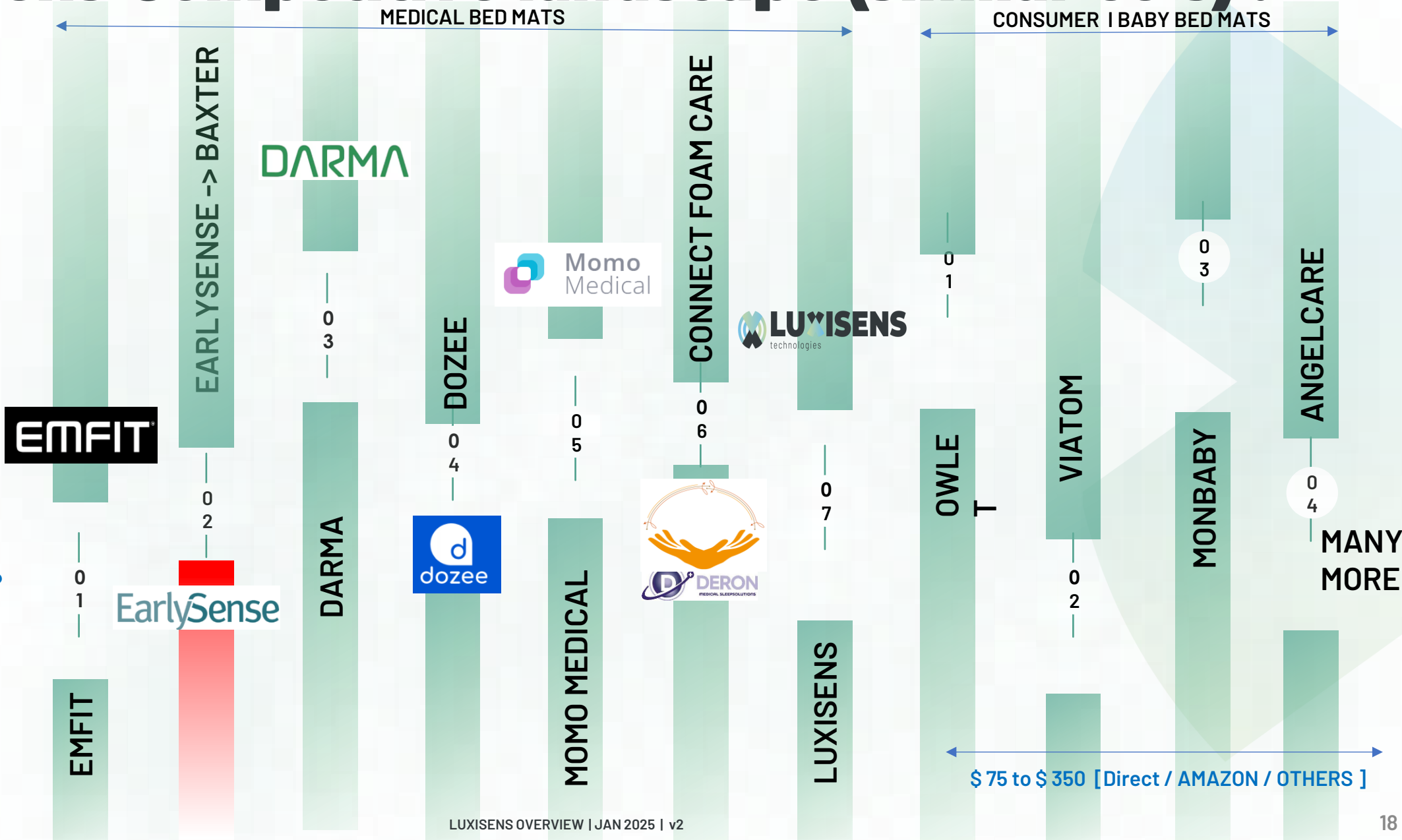
- B2B business: Explore providing advanced algorithms to companies that wants to collaborate.

## 3. New Product Sales (for various Use Cases)

- Establish 1 country direct sales channel (NL).
- Establish Distribution Partners globally.
- Generate Service revenues thru sockets over a period [h/w & s/w].
- Focus: SaaS model. [subscription] – Luxisens COGS is less – we target increasing 'No of Beds' strategy.

# Luxisens Competitive landscape (similar co's) !

- ✓ None are big
- ✓ Expensive
- ✓ Clinically validation ?
- ✓ Accuracy / Reliability ??



# The Regulatory pathway !

Luxisens SensorMat will be classified as Class IIB device [initial analysis]

## UNITED STATES



- Device Common Name:
  - Breathing frequency monitor, Heart rate monitor.
- Classification Name:
  - 21 CFR 868.2375 Breathing frequency monitor
  - 21 CFR 870.2300 Heart rate Monitor
- Regulatory Class: Class II
- Product Code: BZQ/DRT
- Predicate device: Exists [Faster to obtain certification]

## EUROPE



- Classification: Class I / II b (MDR – new guideline)
- Time taking due to established EU notified bodies are not taking new clients.
- Clinical submission compliant to be evaluated as per MDR

Luxisens™ SensorMat™ application usage on people / patients is intended to measure parameters [non-clinical positional data → to→ clinical vital parameters] in general home or in-hospital or clinical setting.

### Indication for Use

- Non-intrusive, continuous measurement of clinical vital parameters to manage patients optimally during sleep or rest conditions.
- Measurement of respiration rate, heart-rate for adults ( <160 kgs) in an automatic manner.
- Providing positional data of human on a bed of either prognostic / diagnostic values.
- The device is intended for use by normal / trained / medical professionals.

Predicate products exists but not using similar Luxisens technology – to start interaction process with regulatory body!



# TEAM

Experienced professionals focused on “Delivering Outcomes & Creating Value”



**Arindam Mukherjee**

CEO

>30 years medtech experience

Responsible for Scale-Up, Innovation  
Strategy  
Driving Business



**Richard Visser**

Operations / Advisor

>30 years medtech experience

Responsible for Manufacturing  
Process, Cost optimization &  
Suppliers



**Henrie van den Boom**

CTO

>30+ years teaching at TU/e

Responsible for Technology, R&D  
& Optical technology  
development.

## Resource Deployment:

- Investment utilization is focused on Product Development in achieving Minimal Viable product [MVP].
- By design: Keeping the Burn-Rate Low till MVP.
- Engaged with external resource: Technology Development Partner & obtaining subsidies.



# Summary

- Luxisens tech is an unique **'IMPACT CREATING'** proposition [SensorMat @ EveryBed].
- **'Un-Obtrusive Monitoring'** is the FUTURE & Luxisens solution fits this emerging trend.
- Mid/Long term monitoring of clinical parameters thru 'Un-Obtrusive' methods will be a clear winner [**clinical accuracy is vital & we have established the 1<sup>st</sup> vital parameter**].
- Market is **broad / big** and higher application capabilities will lead to better traction and hence selection of 'Use Cases' will be critical to create an impact.
- There is **NO** similar or dominant player that commands larger market share. **Consumer** offering will be very competitive but large volumes.
- Providing **connected solution / harnessing data** will increase company value.



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